Sampling with flows, diffusion and autoregressive neural networks: A spin-glass perspective

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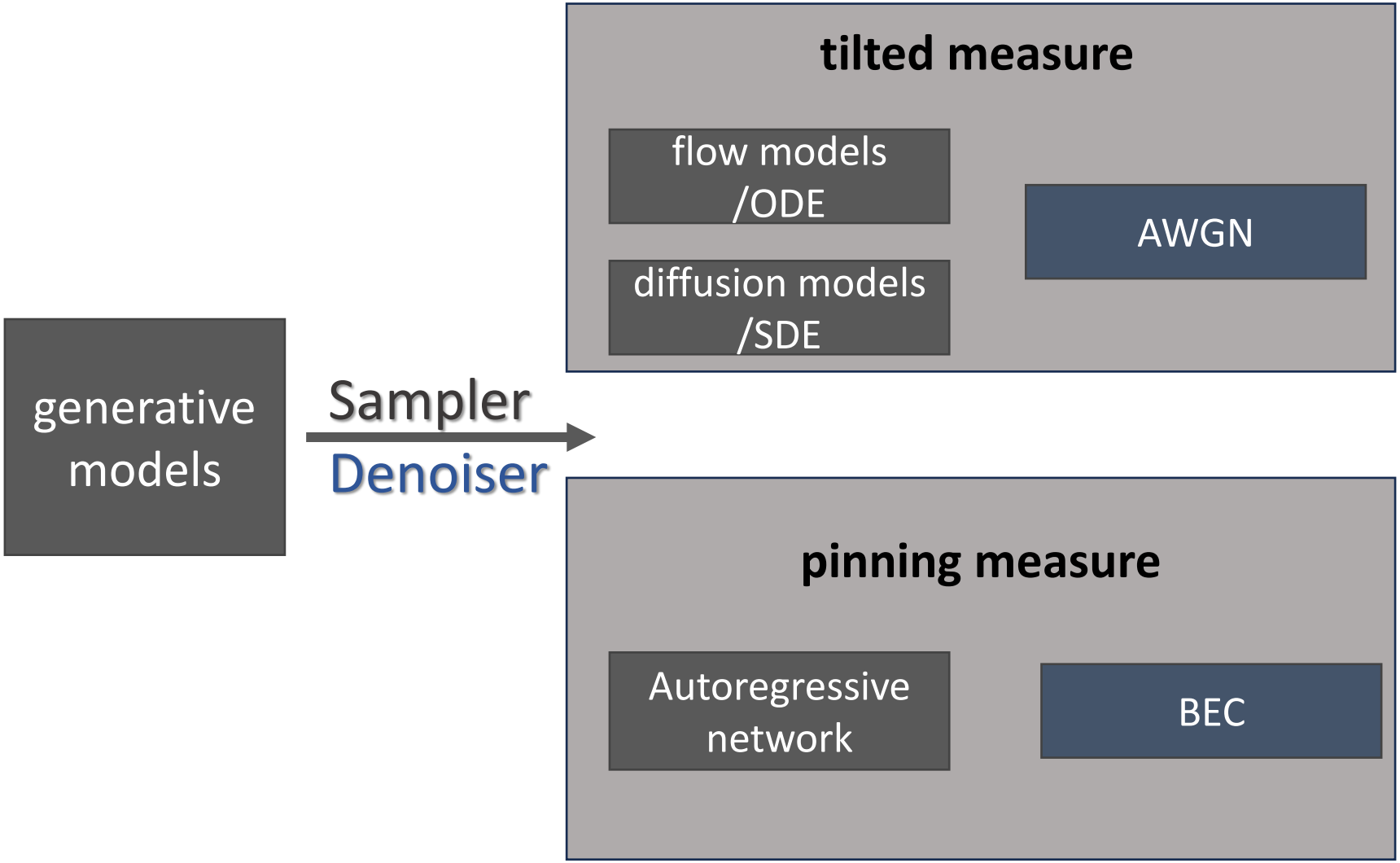
(Sep 23th)

思维导图


tilted measure
flow models /ODE
diffusion models /SDE
AWGN
generative models
Sampler
Denoiser
pinning measure
Autoregressive network
BEC

tilted measure： flow models

模型设定

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbb{I}_N) \qquad \text{sample } \mathbf{x}_0 \sim P_0$$

$$\mathbf{y}(t) = \alpha(t)\mathbf{x}_0 + \beta(t)\mathbf{z} \longrightarrow \begin{array}{c} \alpha(0) = \beta(1) = 1; \quad \alpha(1) = \beta(0) = 0; \\ \forall t \in [0,1] : \alpha(t) \geq 0,\ \dot{\alpha}(t) \leq 0,\ \beta(t) \geq 0,\ \dot{\beta}(t) \geq 0 \end{array}$$

$$\mathbf{Y}_{t=1} = \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbb{I}_N) \longrightarrow \begin{array}{l} \mathbf{Y}_{t=1} = \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbb{I}_N) \\ P_0(\mathbf{x}_0) \text{ at time } t = 0 \end{array}$$

演化方程

$$\partial_t \rho(\mathbf{y}, t) + \nabla \cdot (b(\mathbf{y}, t)\rho(\mathbf{y}, t)) = 0$$

$$b(\mathbf{y}, t) = \mathbb{E}[\partial_t \mathbf{y}(t) | \mathbf{y}(t) = \mathbf{y}] = \mathbb{E}[\dot{\alpha}(t)\mathbf{x_0} + \dot{\beta}(t)z | \mathbf{y}(t) = \mathbf{y}]$$

$$\frac{d\mathbf{Y}(t)}{dt} = b(\mathbf{Y}(t), t) \qquad \mathbf{Y}_{t-\delta} = \mathbf{Y}_t - \delta\, b(\mathbf{Y}_t, t)$$

tilted measure： flow models

出发点

$$b(\mathbf{y},t) = \mathbb{E}[\partial_t \mathbf{y}(t)|\mathbf{y}(t)=\mathbf{y}] = \mathbb{E}[\dot{\alpha}(t)\mathbf{x_0} + \dot{\beta}(t)z|\mathbf{y}(t)=\mathbf{y}]$$

$$\mathbf{Y}_{t-\delta} = \mathbf{Y}_t - \delta\, b(\mathbf{Y}_t, t)$$

$$\begin{aligned} b(\mathbf{Y}_t,t) &= \mathbb{E}\left[\dot{\alpha}(t)\mathbf{x}_0 + \frac{\dot{\beta}(t)}{\beta(t)}\beta(t)\mathbf{z}\middle|\mathbf{y}(t)=\mathbf{Y}_t\right] \\ &= \mathbb{E}\left[\dot{\alpha}(t)\mathbf{x}_0 + \frac{\dot{\beta}(t)}{\beta(t)}\left(\mathbf{y}(t)-\alpha(t)\mathbf{x}_0\right)\middle|\mathbf{y}(t)=\mathbf{Y}_t\right] \\ &= \frac{\dot{\beta}(t)}{\beta(t)}\mathbf{Y}_t + \left(\dot{\alpha}(t) - \frac{\dot{\beta}(t)\alpha(t)}{\beta(t)}\right)\mathbb{E}\left[\mathbf{x}_0|\mathbf{y}(t)=\mathbf{Y}_t\right] \end{aligned}$$

$$\mathbf{Y}_{t-\delta} = \left(1-\delta\frac{\dot{\beta}(t)}{\beta(t)}\right)\mathbf{Y}_t - \delta\left(\dot{\alpha}(t) - \frac{\dot{\beta}(t)\alpha(t)}{\beta(t)}\right)\mathbb{E}\left[\mathbf{x}_0|\mathbf{y}(t)=\mathbf{Y}_t\right]$$

tilted measure： flow models

Algorithm 1 Flow-based sampling algorithm

Input: Denoiser, parameters: $\delta, \alpha(t), \beta(t)$
$\mathbf{Y}_{t=1} = \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbb{I}_N)$, $N_{\text{steps}} \equiv \lfloor 1/\delta \rfloor$
for $l = 0, 1, \ldots, N_{\text{steps}} - 1$ **do**
  $t = 1 - l\delta$; $\alpha = \alpha(t)$; $\beta = \beta(t)$; $\gamma = \alpha/\beta$
  Compute $\hat{\mathbf{x}}_0 = \mathbb{E}\left[\mathbf{x}_0 | \mathbf{y}(t) = \mathbf{Y}_t\right]$ using the denoiser
  (assuming a channel $\mathbf{Y}_t = \alpha \mathbf{X}_0 + \beta \mathbf{Z}$)
  Update the field $\boldsymbol{Y}_{t-\delta}$ via Eq. (9)
end for
Return $\hat{\boldsymbol{x}}_0$

tilted measure： AWGN

目标： $\mathbb{E}\left[\mathbf{x}_0 | \mathbf{y}(t) = \mathbf{Y}_t\right]$

$$P(\mathbf{x}|\mathbf{y}(t) = \mathbf{Y}_t) = \frac{P_0(\mathbf{x}) P(\mathbf{y}(t) = \mathbf{Y}_t|\mathbf{x})}{P(\mathbf{y}(t) = \mathbf{Y}_t)}$$

$$= \frac{1}{Z(\mathbf{Y}_t)} \exp\left(\frac{\alpha(t)}{\beta(t)^2}\langle \mathbf{Y}_t, \mathbf{x}\rangle - \frac{\alpha(t)^2}{2\beta(t)^2}||\mathbf{x}||^2\right) P_0(\mathbf{x})$$

$$\mathbf{y}(t) = \alpha(t)\mathbf{x}_0 + \beta(t)\mathbf{z}$$

$$P_\gamma(\mathbf{x}|\mathbf{x}_0, \mathbf{z}) \propto e^{\gamma(t)^2\langle \mathbf{x}, \mathbf{x}_0\rangle + \gamma(t)\langle z, \mathbf{x}\rangle - \frac{\gamma(t)^2}{2}\|\mathbf{x}\|^2} P_0(\mathbf{x}) \qquad \gamma(t) = \frac{\alpha(t)}{\beta(t)}$$

pinning measure

sequential sampling 观点

$$P_0(\mathbf{x}) = P_0(x_1)P_0(x_2|x_1)\dots P_0(x_N|x_1,\dots,x_{N-1})$$

Information theoretic 观点

binary erasure channel

(BEC) in which a transmitter sends a bit and the receiver either receives the bit correctly or with some probability 1 − θ receives a message that the bit was not received but “erased” instead.

$$y_i = \begin{cases} [\mathbf{x}_0]_i & \text{with probability } \theta \\ * & \text{otherwise} \end{cases}$$

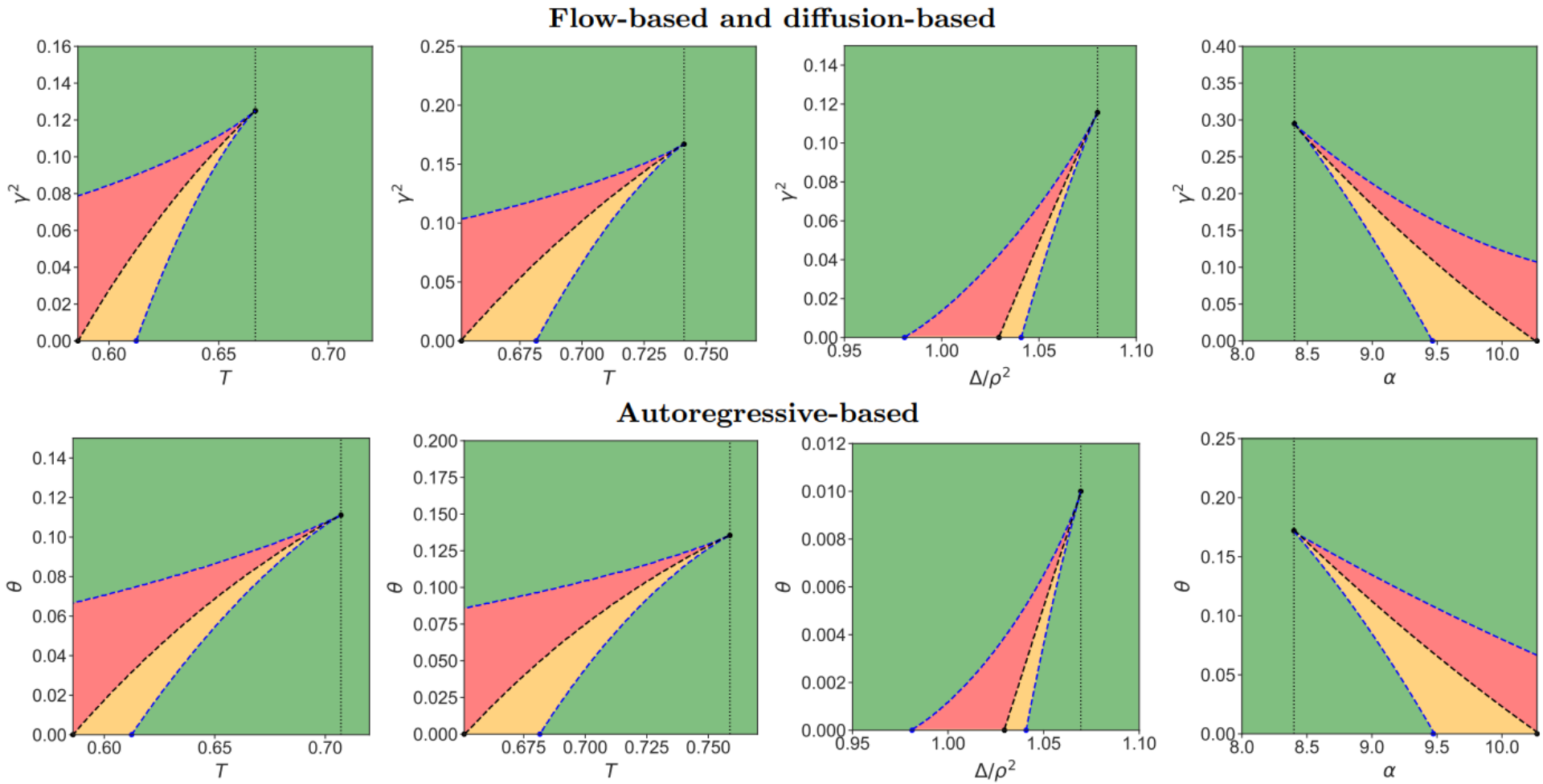
$$P_\theta(\mathbf{x}|\mathbf{x}_0, S_\theta) \propto P_0(\mathbf{x}) \prod_{i \in S_\theta} \delta([x_i - [\mathbf{x}_0]_i)$$

pinning measure

Algorithm 2 Autoregressive-network-based sampling algorithm

Input: BEC denoiser
for $l = 1, \ldots, N$ **do**
  Compute the posterior marginal $P_l(x_l|x_1, \ldots, x_{l-1})$ with BEC denoiser Eq. (15)
  Assign $x_l \sim P_l(x_l|x_1, \ldots, x_{l-1})$
end for
Return $\boldsymbol{x}$

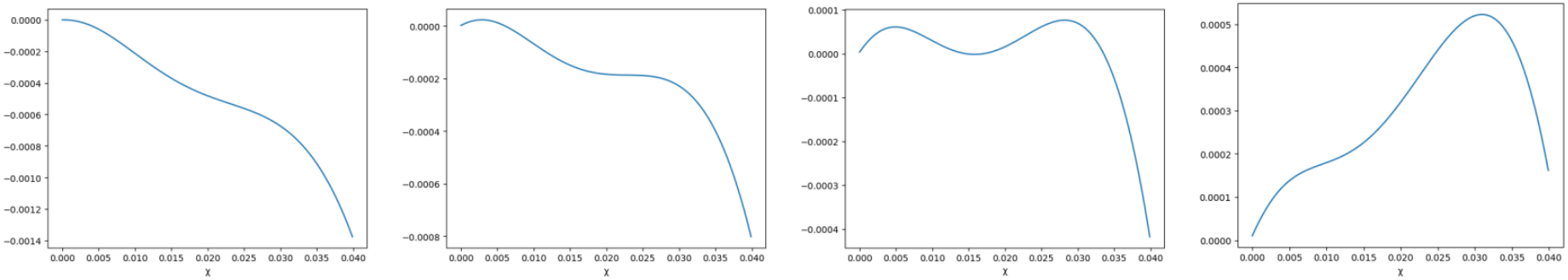
相图



(a) Spherical p-spin (b) Ising p-spin (c) Rank-one estimation (d) Bicoloring

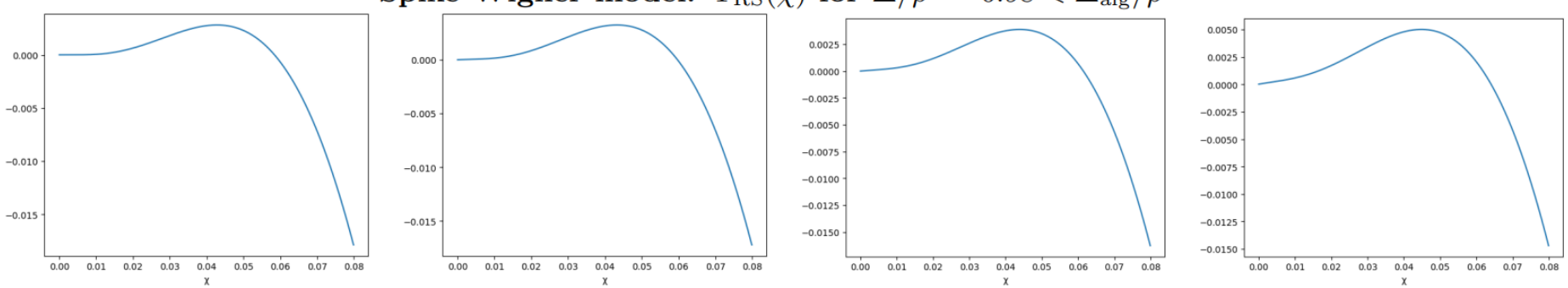
相图

Spike Wigner model: $\Phi_{\mathrm{RS}}(\chi)$ **for** $\Delta_d/\rho^2 < \Delta/\rho^2 = 1.05 < \Delta_{\mathrm{tri}}/\rho^2$



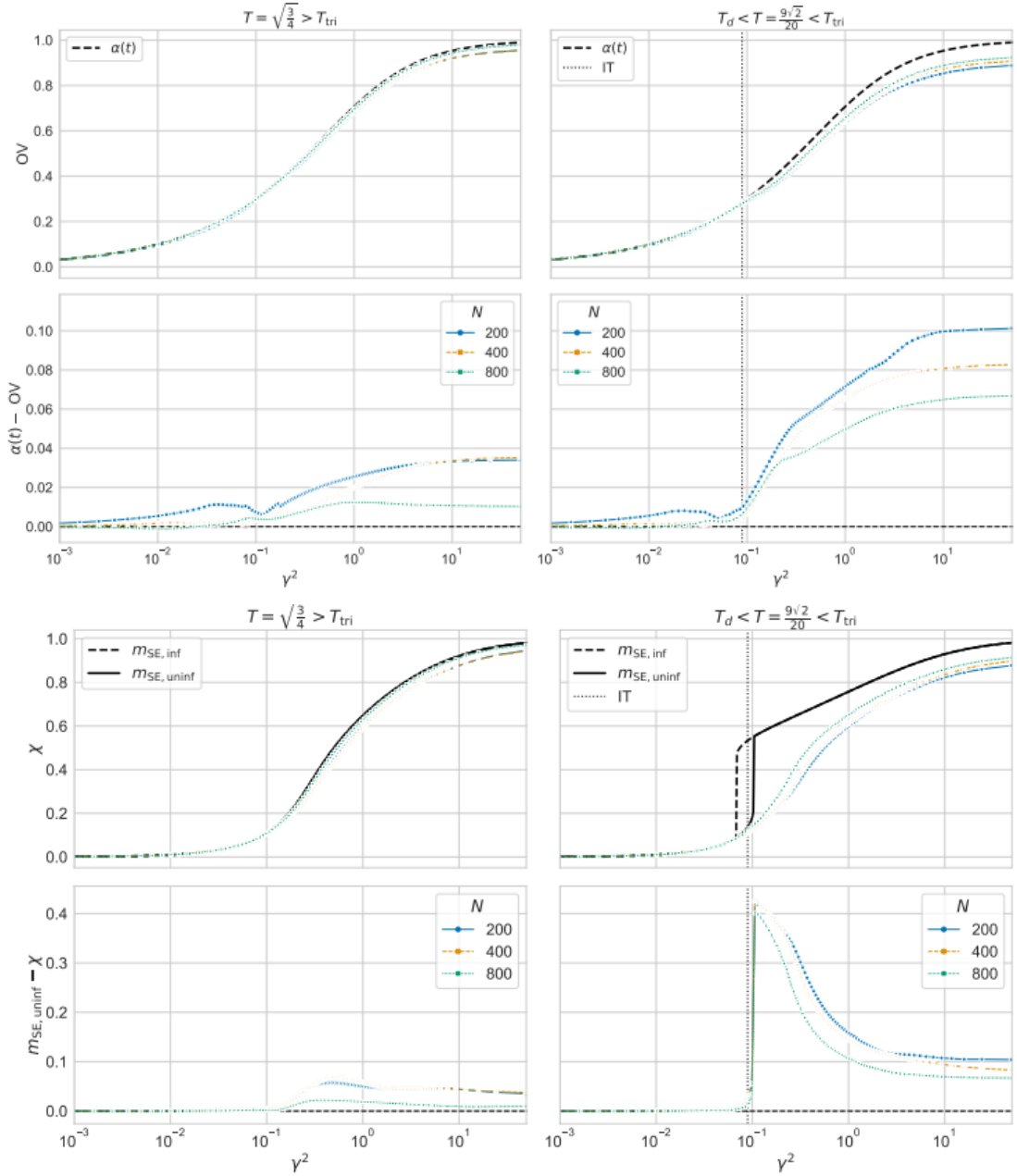
(a) $\gamma^2 = 0$ (b) $\gamma^2 = 0.03$ (c) $\gamma^2 = 0.05$ (d) $\gamma^2 = 0.08$

Spike Wigner model: $\Phi_{\mathrm{RS}}(\chi)$ **for** $\Delta/\rho^2 = 0.98 < \Delta_{\mathrm{alg}}/\rho^2$



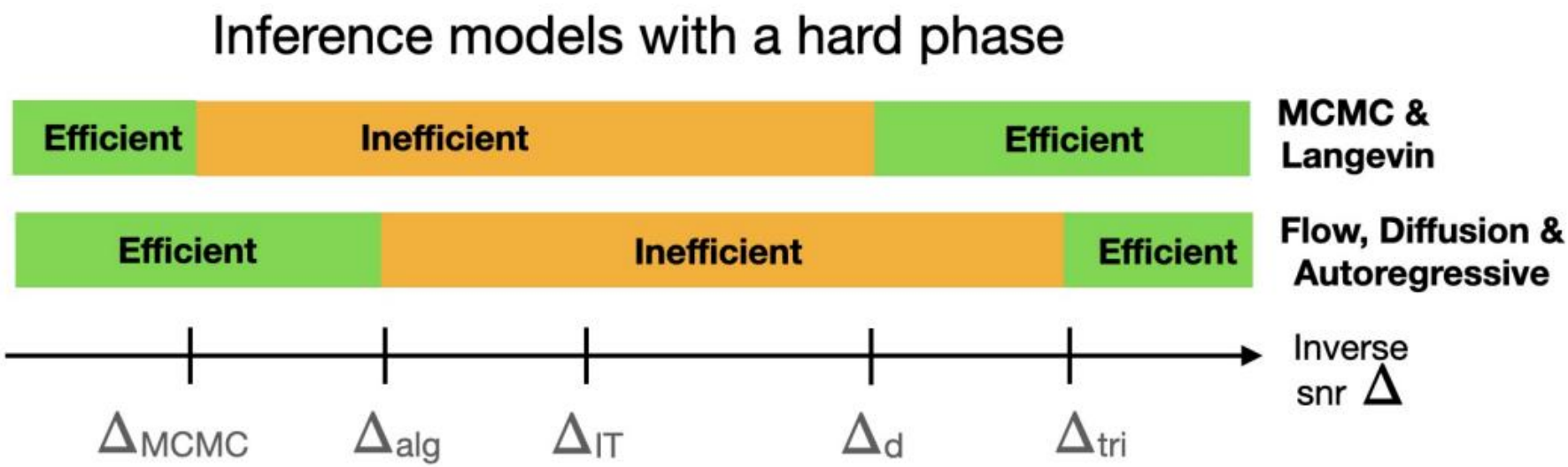
(a) $\gamma^2 = 0$ (b) $\gamma^2 = 0.02$ (c) $\gamma^2 = 0.05$ (d) $\gamma^2 = 0.1$

相图


$T = \sqrt{\frac{3}{4}} > T_{\mathrm{tri}}$
$T_d < T = \frac{9\sqrt{2}}{20} < T_{\mathrm{tri}}$
$\alpha(t)$
IT
OV
$\alpha(t) - \mathrm{OV}$
N
200
400
800
γ^2
$m_{\mathrm{SE,\,inf}}$
$m_{\mathrm{SE,\,uninf}}$
χ
$m_{\mathrm{SE,\,uninf}} - \chi$

相图

Models with random first order transition
Inefficient
Efficient
MCMC & Langevin
Inefficient
Efficient
Flow, Diffusion & Autoregressive
T
T_K
T_d
T_{tri}


Inference models with a hard phase
Efficient
Inefficient
Efficient
MCMC & Langevin
Efficient
Inefficient
Efficient
Flow, Diffusion & Autoregressive
Inverse snr Δ
Δ_{MCMC}
Δ_{alg}
Δ_{IT}
Δ_d
Δ_{tri}

Appendix B

~~Asymptotic solutions and Phase Diagrams~~

Calculation of RS, AMP and SE

September 23, 2023

1. Sparse rank-one matrix factorization

Model setting:

$$W_{ij} = \frac{x_i^\star x_j^\star}{\sqrt{N}} + \tilde{z}_{ij}$$

where $x^\star$ sampled from Rademacher-Bernoulli distribution

$$\boldsymbol{x}^\star \sim P_X(x) = (1-\rho)\delta(x,0) + \frac{\rho}{2}\delta(x,1) + \frac{\rho}{2}\delta(x,-1), \quad \tilde{Z}_{ij} = \tilde{Z}_{ji} \sim \mathcal{N}(0,\Delta)$$

Bayes Rule:

$$P(x|\,W) = \frac{P_X(x)P(\,W|x)}{\int \mathrm{d}x P_X(x) P(\,W|x)}$$

where

$$P(\,W|x) = \frac{1}{Z}\exp\left[-\frac{1}{2\Delta}\sum_{i<j}\left(W_{ij} - \frac{x_i x_j}{\sqrt{N}}\right)^2\right] = \frac{1}{Z}\exp\left[-\frac{1}{2\Delta}\sum_{i<j}\left(\frac{x_i^\star x_j^\star}{\sqrt{N}} + \tilde{z}_{ij} - \frac{x_i x_j}{\sqrt{N}}\right)^2\right]$$

$$\propto \exp\left[-\frac{1}{\Delta}\sum_{i<j}\left(\frac{x_i^2 x_j^2}{2N} - \frac{x_i x_j x_i^\star x_j^\star}{N} - \frac{x_i x_j \tilde{z}_{ij}}{\sqrt{N}}\right)^2\right]$$

define order parameters $\mu \equiv \frac{1}{N}\mathbb{E}\left[\boldsymbol{x}\boldsymbol{x}^\star\right] = m \equiv \frac{1}{N}\mathbb{E}\left[||\boldsymbol{x}||^2\right]$

1.1 replica free entropy

The replica formula for such problems can be found in many places and we refer to the mathematical literature for the detailed rigorous statements: [81, 84, 103, 104, 108, 110]. In particular, [103] gives a generic proof using the adaptive interpolation methods in the presence of a Gaussian channel is added, which turns out to give the same measure as the tilted one, while [104] uses instead a pinned measure. In both case, we can thus adapt the results in the literature: The asymptotic free entropy is given by the maximum of the so-called replica symmetric potential[20] :

$$\frac{1}{N}\mathbb{E}_{\mathbf{x}_0,\mathbf{z},\tilde{\mathbf{z}}} \log Z_\gamma \xrightarrow[N\to\infty]{} \operatorname{argmax} \Phi_{RS}(m) \tag{B7}$$

$$\Phi_{RS}(m) = \mathbb{E}_{w,x_0}\left[\log Z_x\left(\frac{m}{\Delta}, \frac{m}{\Delta}x_0 + \sqrt{\frac{m}{\Delta}}w\right)\right] - \frac{m^2}{4\Delta} \tag{B8}$$

[1] T. Lesieur, F. Krzakala, and L. Zdeborová. Mmse of probabilistic low-rank matrix estimation: Universality with respect to the output channel. arXiv:1507.03857, 2015.

- **Bethe mutual information**

$$i_B(m) = \frac{m^2 + \left[\mathbb{E}_x\left(x^2\right)\right]^2}{4\Delta} - \mathbb{E}_{x,Z}\left[\mathcal{J}\left(\frac{m}{\Delta}, \frac{mx}{\Delta} + \sqrt{\frac{m}{\Delta}}z\right)\right]$$

where

$$\mathcal{J}(A,B) = \log \int \mathrm{d}x P(x) \exp\left(Bx - \frac{1}{2}Ax^2\right)$$

1.1 replica free entropy

[2] Florent Krzakala, Jiaming Xu, and Lenka Zdeborovťa. Mutual information in rank-one matrix estimation. In 2016 IEEE Information Theory Workshop (ITW). IEEE, 2016.

- **Mutual information and free energy**

$$\frac{I(x;\,Y)}{n} = f_n + \frac{\left[\mathbb{E}_x\left(x^2\right)\right]^2}{4\Delta}$$

where

$$I(x;\,Y) = \mathbb{E}_Y[-\log P(Y)] - \mathbb{E}_{x,y}\left[\log\frac{P(x,\,Y)}{P(x)}\right], \quad f_n = -\frac{1}{n}\mathbb{E}_Y[\log Z(Y)]$$

[3] Mohamad Dia, Nicolas Macris, Florent Krzakala, Thibault Lesieur, Lenka Zdeborovťa, et al. Mutual information for symmetric rank-one matrix estimation: A proof of the replica formula. Advances in Neural Information Processing Systems, 29, 2016.

- $$\lim_{n\to\infty}\frac{I(x;\,Y)}{n} = \min_m i_B(m)$$

for (a) $\Delta > 0$

(b) P_0 is a discrete distribution

(c) $i_B(m)$ has at most three stationary points

1.1 replica free entropy

$$-\frac{1}{n}\mathbb{E}[\log Z] = f_n = \frac{I(x;\,Y)}{n} - \frac{\left[\mathbb{E}_x\left(x^2\right)\right]^2}{4\Delta} \xrightarrow{n\to\infty} \min_m i_B(m) - \frac{\left[\mathbb{E}_x\left(x^2\right)\right]^2}{4\Delta}$$

$$= \min_m \frac{m^2}{4\Delta} - \mathbb{E}_{x,Z}\left[\mathcal{J}\left(\frac{m}{\Delta}, \frac{mx}{\Delta} + \sqrt{\frac{m}{\Delta}}z\right)\right]$$

Finally we get

$$\frac{1}{n}\mathbb{E}[\log Z] = \max_m \underbrace{\mathbb{E}_{x_0,z}\left[\mathcal{J}\left(\frac{m}{\Delta}, \frac{mx_0}{\Delta} + \sqrt{\frac{m}{\Delta}}z\right)\right] - \frac{m^2}{4\Delta}}_{\Phi_{\mathrm{RS}}}$$

where $\mathcal{J}(A, B) = \log \underbrace{\int \mathrm{d}x P(x) \exp\left(Bx - \frac{1}{2}Ax^2\right)}_{Z}$

$$\frac{1}{N}\mathbb{E}_{\mathbf{x}_0,\mathbf{z},\tilde{\mathbf{z}}} \log Z_\gamma \xrightarrow[N\to\infty]{} \operatorname{argmax} \Phi_{RS}(m)$$

$$\Phi_{RS}(m) = \mathbb{E}_{w,x_0}\left[\log Z_x\left(\frac{m}{\Delta}, \frac{m}{\Delta}x_0 + \sqrt{\frac{m}{\Delta}}w\right)\right] - \frac{m^2}{4\Delta}$$

1.1 issue in calculation of replica *tilted measure*

- **partition functions**

$$
\begin{aligned}
Z(A,B) &= \int \mathrm{d}x\ P_X(x) \exp\left(\gamma^2 x x_0 + \gamma z x - \frac{1}{2}\gamma^2 x^2\right) \exp\left(Bx - \frac{1}{2}Ax^2\right) \\
&= \int \mathrm{d}x \left[(1-\rho)\delta(x,0) + \frac{\rho}{2}\delta(x,1) + \frac{\rho}{2}\delta(x,-1)\right] \mathrm{e}^{\gamma^2 x x_0 + \gamma z x - \frac{1}{2}\gamma^2 x^2} \mathrm{e}^{Bx - \frac{1}{2}Ax^2} \\
&= (1-\rho) + \rho \exp\left[-\frac{1}{2}\left(A + \gamma^2\right)\right] \cosh\left(B + \gamma z + \gamma^2 x_0\right)
\end{aligned}
$$

- **replica symmetric potential**

$$
\Phi_{\mathrm{RS}}(\mathrm{m}) = \mathbb{E}_{x_0,z}\left[\log Z\left(\frac{m}{\Delta}, \frac{m x_0}{\Delta} + \sqrt{\frac{m}{\Delta}} z\right)\right] - \frac{m^2}{4\Delta}
$$

$$
\begin{aligned}
&\log Z\left(\frac{m}{\Delta}, \frac{m x_0}{\Delta} + \sqrt{\frac{m}{\Delta}} z\right) \\
&= \log\left\{\rho \exp\left[-\frac{1}{2}\left(\frac{m}{\Delta} + \gamma^2\right)\right] \cosh\left[\left(\frac{m}{\Delta} + \gamma^2\right) x_0 + \left(\sqrt{\frac{m}{\Delta}} + \gamma\right) z\right] + (1-\rho)\right\} \\
&= \log\left\{\rho \exp\left(-\frac{1}{2}\widetilde{m}\right) \cosh\left(\widetilde{m} x_0 + \sqrt{\widetilde{m}} z\right) + (1-\rho)\right\}
\end{aligned}
$$

define $\widetilde{m} = \frac{m}{\Delta} + \gamma^2$, but $\sqrt{\widetilde{m}} = \sqrt{\frac{m}{\Delta} + \gamma^2} \neq \sqrt{\frac{m}{\Delta}} + \gamma$

1.1 issue in calculation of replica *tilted measure*

$$
\begin{aligned}
\mathbb{E}_z\ \mathbb{E}_{x_0}[\log Z] =& (1-\rho)\log\left[(1-\rho)+\rho \mathrm{e}^{-\frac{\widetilde{m}}{2}}\cosh(\sqrt{\widetilde{m}}z)\right] \\
&+\frac{\rho}{2}\log\left[(1-\rho)+\rho \mathrm{e}^{-\frac{\widetilde{m}}{2}}\cosh(\widetilde{m}+\sqrt{\widetilde{m}}z)\right] \\
&+\frac{\rho}{2}\log\left[(1-\rho)+\rho \mathrm{e}^{-\frac{\widetilde{m}}{2}}\cosh(-\widetilde{m}+\sqrt{\widetilde{m}}z)\right]-\frac{m^2}{4\Delta}
\end{aligned}
$$

$$
= \mathbb{E}_z(1-\rho)\log\left[(1-\rho)+\rho \mathrm{e}^{-\frac{\widetilde{m}}{2}}\cosh(\sqrt{\widetilde{m}}z)\right]+\rho\log\left[(1-\rho)+\rho \mathrm{e}^{-\frac{\widetilde{m}}{2}}\cosh(\widetilde{m}+\sqrt{\widetilde{m}}z)\right]-\frac{m^2}{4\Delta}
$$

1.1 issue in calculation of replica *pinning measure*

- **partition functions**

$$Z(A,B) = \begin{cases} \int \mathrm{d}x P_X(x)\delta_{x,x_0} \exp\left(Bx - Ax^2/2\right) & \text{with probability } \theta \\ \int \mathrm{d}x P_X(x) \exp\left(Bx - Ax^2/2\right) & \text{with probability } 1-\theta \end{cases}$$

$$= \begin{cases} P_X(x_0) \exp\left(Bx_0 - Ax_0^2/2\right) & \text{with probability } \theta \\ \rho \exp(-A/2)\cosh(B) + (1-\rho) & \text{with probability } 1-\theta \end{cases}$$

- **replica symmetric potential**

$$\begin{aligned} \Phi_{\mathrm{RS}}(m) =& \theta\left(\rho\log\rho + (1-\rho)\log(1-\rho) + \frac{\rho}{2}\tilde{m}\right) \\ &+ (1-\theta)\left[\rho\mathbb{E}_w\left[\log\left((1-\rho) + \rho e^{-\tilde{m}/2}\cosh(\tilde{m} + \sqrt{\tilde{m}}w)\right)\right]\right. \\ &\left.+(1-\rho)\mathbb{E}_w\left[\log\left((1-\rho) + \rho e^{-\tilde{m}/2}\cosh(\sqrt{\tilde{m}}w)\right)\right]\right] - \frac{m^2}{4\Delta} \end{aligned}$$

define $\tilde{m} = \frac{m}{\Delta}$

1.1 issue in calculation of replica *pinning measure*

$$\Phi_{\mathrm{RS}}(\mathrm{m}) = \mathbb{E}_{x_0,z}\left[\log Z\left(\frac{m}{\Delta}, \frac{mx_0}{\Delta} + \sqrt{\frac{m}{\Delta}}z\right)\right] - \frac{m^2}{4\Delta}$$

- **with probability** θ

$$\begin{aligned}\mathbb{E}_{x_0}[\log Z] =& (1-\rho)\log(1-\rho) \\ &+ \frac{\rho}{2}\log\left[\frac{\rho}{2}\exp\left(\frac{m}{2\Delta}+\sqrt{\frac{m}{\Delta}}z\right)\right] + \frac{\rho}{2}\log\left[\frac{\rho}{2}\exp\left(\frac{m}{2\Delta}-\sqrt{\frac{m}{\Delta}}z\right)\right] \\ =& (1-\rho)\log(1-\rho) + \rho\log\left[\frac{\rho}{2}\exp\left(\frac{m}{2\Delta}\right)\right] \\ =& (1-\rho)\log(1-\rho) + \rho\log\rho + \rho\log\frac{1}{2} + \frac{\rho}{2}\frac{m}{\Delta}\end{aligned}$$

- **with probability** $1-\theta$

$$\frac{\rho}{2}\log[A + B\cosh(C+D)] + \frac{\rho}{2}\log[A + B\cosh(-C+D)] = \rho\log[A + B\cosh(C+D)]$$

1.2 AMP equation

[4] Thibault Lesieur, Florent Krzakala, and Lenka Zdeborovťa. Constrained low-rank matrix estimation: Phase transitions, approximate message passing and applications. Journal of Statistical Mechanics: Theory and Experiment, 2017(7):073403, 2017

- **BP equation**

$$n_{i\to ij}\left(x_i\right) = \frac{1}{Z_{i\to ij}} P_X\left(x_i\right) \prod_{1\le k\le N, k\neq i,j} \tilde{n}_{ki\to i}\left(x_i\right)$$

$$\tilde{n}_{ki\to i}\left(x_i\right) = \frac{1}{Z_{ki\to i}} \int \mathrm{d}x\ n_{k\to ki}\left(x_k\right) \mathrm{e}^{g}$$

- **relax-BP equation** expanded to the second order

$$e^{g\left(Y_{ij}, w_{ij}\right)} = e^{g\left(Y_{ij}, 0\right)} \left\{ 1 + \left.\frac{\partial g\left(Y_{ij}, w\right)}{\partial w}\right|_{w=0} w_{ij} + \left[\left(\left.\frac{\partial g\left(Y_{ij}, w\right)}{\partial w}\right|_{w=0} \right)^2 + \left.\frac{\partial^2 g\left(Y_{ij}, w\right)}{\partial w^2}\right|_{w=0} \right] \frac{w_{ij}^2}{2} + \mathcal{O}\left(w_{ij}^3\right) \right\}$$

define

$$S_{ij} = \left.\frac{\partial g\left(Y_{ij}, w\right)}{\partial w}\right|_{w=0} \qquad R_{ij} = \left(\left.\frac{\partial g\left(Y_{ij}, w\right)}{\partial w}\right|_{w=0} \right)^2 + \left.\frac{\partial^2 g\left(Y_{ij}, w\right)}{\partial w^2}\right|_{w=0}$$

1.2 AMP equation

define mean and variance

$$\hat{x}_{k\to ki} = \int \mathrm{d}x_k n_{k\to ki}\left(x_k\right) x_k \qquad \sigma_{x,k\to ki} = \int \mathrm{d}x_k n_{k\to ki}\left(x_k\right) x_k^2 - \hat{x}_{k\to ki}^2$$

rewrite BP equation

$$\tilde{n}_{ki\to i}\left(x_i\right) = \frac{1}{Z_{ki\to i}}$$

$$\exp\left[g\left(Y_{ki},0\right) + \frac{S_{ki}\hat{x}_{k\to ki}}{\sqrt{N}} x_i - \frac{S_{ki}^2\hat{x}_{k\to ki}^2}{2N} x_i^2 + \frac{R_{ki}\left(\hat{x}_{k\to ik}^2 + \sigma_{x,k\to ik}\right)}{2N} x_i^2 + O\left(\frac{1}{N^{3/2}}\right)\right]$$

define

$$B_{X,i\to ij} = \frac{1}{\sqrt{N}} \sum_{\substack{1\le k\le N\\ k\neq j}} S_{ki}\hat{x}_{k\to ki}, \qquad A_{X,i\to ij} = \frac{1}{N} \sum_{\substack{1\le k\le N\\ k\neq j}} \left[S_{ki}^2\hat{x}_{k\to ki}^2 + R_{ki}\left(\hat{x}_{k\to ik}^2 + \sigma_{x,k\to ik}\right)\right]$$

so we get

$$n_{i\to ij}\left(x_i\right) = \frac{1}{Z_{i\to ij}} P_X\left(x_i\right) \exp\left(B_{X,i\to ij}x_i - \frac{1}{2}A_{X,i\to ij}x_i^2\right)$$

calculate mean and variance

$$f_{\mathrm{in}}^x\left(A,B\right) \equiv \frac{\int \mathrm{d}x\ xP_X(x)\exp\left(Bx - \frac{Ax^2}{2}\right)}{\int \mathrm{d}xP_X(x)\exp\left(Bx - \frac{Ax^2}{2}\right)} \qquad \frac{\partial f_{\mathrm{in}}^x(A,B)}{\partial B} = \mathbb{E}_{\mathcal{W}(\mathcal{A},\mathcal{B})}\left(x^2\right) - \left[f_{\mathrm{in}}^x(A,B)\right]^2$$

1.2 AMP equation

- **AMP equation** remove $\to$ in r-BP

$$B^t_{X,i} - B^t_{X,i\to ij} = \frac{S_{ij}}{\sqrt{N}} \hat{x}^t_{j\to ij} = O\left(\frac{1}{\sqrt{N}}\right)$$

$$A_{X,i} - A_{X,i\to ij} = O\left(\frac{1}{N}\right)$$

AMP

$$B^t_{X,i} = \frac{1}{\sqrt{N}} \sum_{k=1}^{N} S_{ki} \hat{x}^t_k - \hat{x}^{t-1}_i \frac{1}{N} \sum_{k=1}^{N} S^2_{ki} \sigma^t_{x,k},$$

$$A^t_{X,i} = \frac{1}{N} \sum_{k=1}^{N} \left[S^2_{ki} \hat{x}^{t,2}_k - R_{ki} \left(\hat{x}^{t,2}_k + \sigma^t_{x,k} \right) \right]$$

1.2 AMP equation

the only difference is the different measures

$$f_{\text{in}}^{x}\left(A,B\right) \equiv \frac{\int \mathrm{d}x P_X(x) \exp\left(B^\top x - \frac{x^\top A x}{2}\right) x}{\int \mathrm{d}x P_X(x) \exp\left(B^\top x - \frac{x^\top A x}{2}\right)}$$

tilted measure

$$f_{\text{in}}\left(A,B\right) = \frac{\int \mathrm{d}x x P_X(x) \exp\left(\gamma^2 x x_0 + \gamma w x - \gamma^2 x^2/2\right) \exp\left(Bx - Ax^2/2\right)}{\int \mathrm{d}x P_X(x) \exp\left(\gamma^2 x x_0 + \gamma w x - \gamma^2 x^2/2\right) \exp\left(Bx - Ax^2/2\right)}$$

pining measure

$$f_{\text{in}}\left(A,B\right) = \begin{cases} \frac{P_X(x_0) x_0 \exp\left(Bx_0 - Ax_0^2/2\right)}{P_X(x_0) \exp\left(Bx_0 - Ax_0^2/2\right)} & \text{with probability } \theta \\ \frac{\int \mathrm{d}x x P_X(x) \exp\left(Bx - Ax^2/2\right)}{\int \mathrm{d}x P_X(x) \exp\left(Bx - Ax^2/2\right)} & \text{with probability } 1-\theta \end{cases}$$

1.2 AMP equation

AMP for tilted measure

$$
\begin{cases}
\widehat{x}_i^{t+1} = \dfrac{\rho \tanh\left(B_i^t + \frac{\alpha(t)}{\beta(t)^2}[\mathbf{Y}_t]_i\right)}{\rho + \frac{(1-\rho)\exp\left(\left(A^t+\gamma^2\right)/2\right)}{\cosh\left(B_i^t + \frac{\alpha(t)}{\beta(t)^2}[\mathbf{Y}_t]_i\right)}} \\
\sigma_i^{t+1} = \rho \dfrac{\rho + (1-\rho)e^{\left(A^t+\gamma^2\right)/2}\cosh\left(B_i^t + \frac{\alpha(t)}{\beta(t)^2}[\mathbf{Y}_t]_i\right)}{\left(\rho\cosh\left(B_i^t + \frac{\alpha(t)}{\beta(t)^2}[\mathbf{Y}_t]_i\right) + (1-\rho)\exp\left(\left(A+\gamma^2\right)/2\right)\right)^2} \\
A^t = \frac{\|\widehat{\mathbf{x}}^{\mathbf{t}}\|_2^2}{\Delta N} \\
B_i^t = \frac{1}{\Delta\sqrt{N}}\mathbf{J_i}\cdot\widehat{\mathbf{x}}^{\mathbf{t}} - \frac{1}{N\Delta}\widehat{x}_i^{t-1}\sum_k \sigma_k^t
\end{cases}
$$

AMP for pinning measure

$$
\begin{cases}
\widehat{x}_i^{t+1} = \begin{cases} [\mathrm{x}_0]_i & \text{if } i \in S_\theta \\ \dfrac{\rho\tanh\left(B_i^t\right)}{\rho + \frac{(1-\rho)\exp\left(A^t/2\right)}{\cosh\left(B_i^t\right)}} & \text{otherwise} \end{cases} \\
\sigma_i^{t+1} = \begin{cases} 0 & \text{if } i \in S_\theta \\ \rho\dfrac{\rho + (1-\rho)e^{A^t/2}\cosh\left(B_i^t\right)}{\left(\rho\cosh\left(B_i^t\right) + (1-\rho)\exp(A/2)\right)^2} & \text{otherwise} \end{cases} \\
A^t = \frac{\|\widehat{\mathbf{x}}^t\|_2^2}{\Delta N} \\
B_i^t = \frac{1}{\Delta\sqrt{N}}\mathbf{J_i}\cdot\widehat{\mathbf{x}}^{\mathrm{t}} - \frac{1}{N\Delta}\widehat{x}_i^{t-1}\sum_k \sigma_k^t
\end{cases}
$$

1.3 SE equation

[5] Thibault Lesieur, Florent Krzakala, and Lenka Zdeborovťa. Constrained low-rank matrix estimation: Phase transitions, approximate message passing and applications. Journal of Statistical Mechanics: Theory and Experiment, 2017(7):073403, 2017

consider Bayes-optimal, defined order parameter

$$M_x^t = \frac{1}{N} \sum_{1 \le i \le N} \hat{x}_i^t x_i^0 \quad Q_x^t = \frac{1}{N} \sum_{1 \le i \le N} \hat{x}_i^t x_i^t$$

$M_x^t = Q_x^t$ under Nishimori condition

take $N \to \infty$, under the central limit theorem

$$\left.\begin{aligned} \mathbb{E}\left(B_{X,i}^t\right) &= \tfrac{M_x^t}{\Delta} x_i^0 \\ \mathbb{D}\left(B_{X,i}^t\right) &= \tfrac{Q_x^t}{\Delta} \\ \mathbb{E}\left(B_{X,i}^t\right) &= \tfrac{Q_x^t}{\Delta} \end{aligned}\right\} M_x^{t+1} = \mathbb{E}_{x_0, W}\left[f_{\text{in}}^x\left(\frac{Q_x^t}{\Delta}, \frac{M_x^t}{\Delta} x_0 + \sqrt{\frac{Q_x^t}{\Delta}} W\right) x_0^\top\right]$$

1.3 SE equation

SE for tilted measure

$$\chi^{t+1} = \rho^2 \mathbb{E}_w \left[\frac{\tanh\left(\widetilde{\chi}^t + \sqrt{\widetilde{\chi}^t} w\right)}{\rho + \frac{(1-\rho)\exp\left(\widetilde{\chi}^t/2\right)}{\cosh\left(\widetilde{\chi}^t + \sqrt{\widetilde{\chi}^t} w\right)}} \right], \quad \tilde{\chi}^t \equiv \frac{\chi^t}{\Delta} + \gamma^2$$

SE for pinning measure

$$\frac{\chi^{t+1}}{\rho} = \theta + (1-\theta)\mathbb{E}_z \left[\frac{\rho \tanh\left(\widetilde{\chi}^t + \sqrt{\widetilde{\chi}^t} w\right)}{\rho + \frac{(1-\rho)\exp\left(\widetilde{\chi}^t/2\right)}{\cosh\left(\widetilde{\chi}^t + \sqrt{\widetilde{\chi}^t} w\right)}} \right], \quad \tilde{\chi}^t \equiv \frac{\chi^t}{\Delta}$$

2. Ising p-spin model & Spherical p-spin model

Model setting:

$$\mathcal{H}(\mathbf{x}) = -\frac{\sqrt{3}}{N} \sum_{i<j<k} J_{ijk} x_i x_j x_k$$

where $J_{ijk} \sim \mathcal{N}(0,1)$
for Ising p-spin model $P_X(x) = \frac{1}{2}\delta(x,-1) + \frac{1}{2}\delta(x,+1)$
for Spherical p-spin model $P_X(x) = \mathcal{N}(0,1)$

$$P_0(\boldsymbol{x}) \propto \prod_i P_X(x_i) \prod_{i<j<k} \mathrm{e}^{\frac{\sqrt{3}\beta}{N} J_{ijk} x_i x_j x_k}$$

[6] Thibault Lesieur, Lťeo Miolane, Marc Lelarge, Florent Krzakala, and Lenka Zdeborovťa. Statistical and computational phase transitions in spiked tensor estimation. In 2017 IEEE International Symposium on Information Theory (ISIT), pages 511515, 2017.

$$\frac{1}{N}\mathbb{E}_{\mathbf{x}_0,\mathbf{z},J} \log Z_\gamma \underset{N\to\infty}{\longrightarrow} \operatorname{argmax} \Phi_{RS}(m)$$

$$\Phi_{RS}(m) = \mathbb{E}_{w,x_0}\left[\log Z_x\left(\frac{m^2}{\Delta}, \frac{m^2}{\Delta}x_0 + \sqrt{\frac{m^2}{\Delta}}w\right)\right] - \frac{m^3}{3\Delta}$$

4. NAE-SAT model

k-hypergraph bicoloring problem $\Longleftrightarrow$ k-NAE-SAT problem

- **k-NAE-SAT**

k-SAT: Boolean Satisfiability Problem

NAE: Not all equal

$$(x_1 \vee x_2 \vee x_3) \wedge (\neg x_1 \vee \neg x_2 \vee \neg x_3)$$

- **k-hypergraph bicoloring**

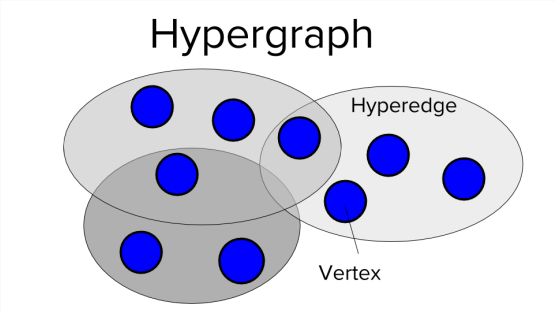
an hypergraph G

with N vertices and M hyperedges

define $\alpha = M/N$

each hyperedge contains exactly k vertices

at least one $+1$ and one -1 among the k variables of $\boldsymbol{x}_{\partial a}$



$$P_0(\boldsymbol{x}) = \frac{1}{Z(G)} \prod_{a=1}^{M} \omega(\boldsymbol{x}_{\partial a}), \quad \omega\left(x_1, \ldots, x_k\right) = \begin{cases} 0 & \text{if } \sum_{i=1}^{k} x_i = \pm k \\ 1 & \text{otherwise} \end{cases}$$

4.1 BP equations and Bethe free-energy

BP equations

$$\eta_{i\to a}(\sigma_i) = \frac{1}{z_0^{ia}} \prod_{b\in\partial i\setminus a} \widehat{\eta}_{b\to i}(\sigma_i)$$

$$\widehat{\eta}_{a\to i}(\sigma_i) = \frac{1}{\widehat{z}_0^{ai}} \sum_{\underline{\sigma}_{\partial a\setminus i}} \omega(\underline{\sigma}_{\partial a}) \prod_{j\in\partial a\setminus i} \eta_{j\to a}(\sigma_j)$$

[6] Louise Budzynski, Federico Ricci-Tersenghi, and Guilhem Semerjian. Biased landscapes for random constraint satis- faction problems. Journal of Statistical Mechanics: Theory and Experiment, 2019(2):023302, feb 2019.

consider $\sigma_i = \pm 1$, refer to SMNN chapter 2 & 3

$$\eta_{i\to a} = \frac{1 + h_{i\to a}\sigma_i}{2} \qquad \widehat{\eta}_{a\to i}(\sigma_i) = \frac{1 + u_{i\to a}\sigma_i}{2}$$

rewrite BP equation

$$h_{i\to a} = f\left(\{u_{b\to i}\}_{b\in\partial_i\setminus a}\right), \qquad u_{a\to i} = g\left(\{h_{j\to a}\}_{j\in\partial_a\setminus i}\right)$$

where

$$f(u_1,\ldots,u_d) = \frac{\prod_{i=1}^d (1+u_i) - \prod_{i=1}^d (1-u_i)}{\prod_{i=1}^d (1+u_i) + \prod_{i=1}^d (1-u_i)}$$

$$g(h_1,\ldots,h_{k-1}) = \frac{\sum_{x_1,\ldots,x_k} \omega(x_1,\ldots,x_k)\, x_k \prod_{i=1}^{k-1}(1+h_i x_i)}{\sum_{x_1,\ldots,x_k} \omega(x_1,\ldots,x_k) \prod_{i=1}^{k-1}(1+h_i x_i)}$$

4.1 BP equations and Bethe free-energy

computational free energy

$$\frac{1}{N}\ln Z(G) = \frac{1}{N}\sum_{i=1}^{N}\ln \mathcal{Z}_0^{\mathrm{v}}\left(\{u_{a\to i}\}_{a\in\partial i}\right) + \frac{1}{N}\sum_{a=1}^{M}\ln \mathcal{Z}_0^{\mathrm{c}}\left(\{h_{i\to a}\}_{i\in\partial a}\right) - \frac{1}{N}\sum_{(i,a)}\ln \mathcal{Z}_0^{\mathrm{e}}\left(h_{i\to a}, u_{a\to i}\right)$$

where

$$\mathcal{Z}_0^{\mathrm{v}}\left(u_1,\ldots,u_d\right) = \sum_{\sigma}\prod_{i=1}^{d}\left(\frac{1+\sigma u_i}{2}\right)$$

$$\mathcal{Z}_0^{\mathrm{c}}\left(h_1,\ldots,h_k\right) = \sum_{\sigma_1,\ldots,\sigma_k}\omega\left(\sigma_1,\ldots,\sigma_k\right)\prod_{i=1}^{k}\left(\frac{1+\sigma_i h_i}{2}\right)$$

$$\mathcal{Z}_0^{\mathrm{e}}(h,u) = \sum_{\sigma}\left(\frac{1+\sigma h}{2}\right)\left(\frac{1+\sigma u}{2}\right)$$

4.2 1RSB and Tree reconstruction equations

When the density of interactions α becomes large, the hypothesis underlying the Replica Symmetric assumption breaks down, and we have to consider the Replica Symmetry Breaking (RSB) phenomenon [119].

c. *1RSB and Tree reconstruction equations:*

The 1RSB cavity method aims to compute the potential

$$\Phi_1(m) = \lim_{N\to\infty} \frac{1}{N} \log \left(\sum_\gamma Z_\gamma^m \right) \tag{B64}$$

where m is the so-called Parisi parameter.

In their general formulation, the 1RSB equations are self-consistent equations defined on distributions over probability distributions. Focusing on the case $m = 1$ allows simplifying considerably the equations, and the resulting formulas correspond to the so-called *Tree Reconstruction* equations [120]. Moreover, the problem we are

index with γ the partition of the configuration space

$$\Phi_1(m) = \lim_{N\to\infty} \frac{1}{N} \ln \left(\sum_\gamma (Z_\gamma)^m \right)$$

4.2.1 1RSB cavity equations

two distributions $P_{i\to a}$ and $\widehat{P}_{a\to i}$, encode the laws of $h^{\gamma}_{i\to a}$ and $u^{\gamma}_{a\to i}$ when the pure state γ is chosen randomly with a probability proportional to Z^m_γ

$$P_{i\to a} = F\left(\left\{\widehat{P}_{b\to i}\right\}_{b\in\partial i\setminus a}\right), \quad \widehat{P}_{a\to i} = G\left(\left\{P_{j\to a}\right\}_{j\in\partial a\setminus i}\right)$$

where

$$P(h) = \frac{1}{z_1\left(\widehat{P}_1,\ldots,\widehat{P}_d\right)}\int\left(\prod_{i=1}^{d}\mathrm{d}u_i\widehat{P}_i\left(u_i\right)\right)\delta\left(h-f\left(u_1,\ldots,u_d\right)\right)z_0\left(u_1,\ldots,u_d\right)^m$$

$$\widehat{P}(u) = \frac{1}{\widehat{z}_1\left(P_1,\ldots,P_{k-1}\right)}\int\left(\prod_{i=1}^{k-1}\mathrm{d}h_iP_i\left(h_i\right)\right)\delta\left(u-g\left(h_1,\ldots,h_{k-1}\right)\right)\widehat{z}_0\left(h_1,\ldots,h_{k-1}\right)^m$$

define

$$\mathcal{P}^{\mathrm{1RSB}}(P) = \sum_{d=0}^{\infty}p_d\int\left(\prod_{i=1}^{d}\mathrm{d}\widehat{P}_i\widehat{\mathcal{P}}^{\mathrm{1RSB}}\left(\widehat{P}_i\right)\right)\delta\left[P-F\left(\widehat{P}_1,\ldots,\widehat{P}_d\right)\right]$$

$$\widehat{\mathcal{P}}^{\mathrm{1RSB}}(\widehat{P}) = \int\left(\prod_{i=1}^{k-1}\mathrm{d}P_i\mathcal{P}^{\mathrm{1RSB}}\left(P_i\right)\right)\delta\left[\widehat{P}-G\left(P_1,\ldots,P_{k-1}\right)\right]$$

4.2.1 1RSB cavity equations

The 1RSB potential

$$\Phi_1(m) = \sum_{d=0}^{\infty} p_d \int \left(\prod_{i=1}^{d} \mathrm{d}\widehat{P}_i \widehat{P}^{\mathrm{1RSB}} \left(\widehat{P}_i \right) \right) \ln \mathcal{Z}_1^{\mathrm{v}} \left(\widehat{P}_1, \ldots, \widehat{P}_d \right)$$
$$+ \alpha \int \left(\prod_{i=1}^{k} \mathrm{d}P_i \mathcal{P}^{\mathrm{1RSB}} \left(P_i \right) \right) \ln \mathcal{Z}_1^{\mathrm{c}} \left(P_1, \ldots, P_k \right)$$
$$- \alpha k \int \mathrm{d}P \, \mathrm{d}\widehat{P} \mathcal{P}^{\mathrm{1RSB}}(P) \widehat{\mathcal{P}}^{\mathrm{1RSB}}(\widehat{P}) \ln \mathcal{Z}_1^{\mathrm{e}}(P, \widehat{P}),$$

where

$$\mathcal{Z}_1^{\mathrm{v}} \left(\widehat{P}_1, \ldots, \widehat{P}_d \right) = \int \left(\prod_{i=1}^{d} \mathrm{d}u_i \widehat{P}_i \left(u_i \right) \right) \left(\mathcal{Z}_0^{\mathrm{v}} \left(u_1, \ldots, u_d \right) \right)^m$$

$$\mathcal{Z}_1^{\mathrm{c}} \left(P_1, \ldots, P_k \right) = \int \left(\prod_{i=1}^{k} \mathrm{d}h_i P_i \left(h_i \right) \right) \left(\mathcal{Z}_0^{\mathrm{c}} \left(h_1, \ldots, h_k \right) \right)^m$$

$$\mathcal{Z}_1^{\mathrm{e}}(P, \widehat{P}) = \int \mathrm{d}h \, \mathrm{d}u P(h) \widehat{P}(u) \left(\mathcal{Z}_0^{\mathrm{e}}(h, u) \right)^m$$

4.2.2 Simplifications for $m = 1$

the normalization constant z_1 does not depend on the whole distributions $\widehat{P}_1, \ldots, \widehat{P}_d$, but only on their average values $\int \mathrm{d}u_i \widehat{P}_i(u_i)\, u_i$

$$P(h) = \frac{1}{z_1\left(\widehat{P}_1, \ldots, \widehat{P}_d\right)} \int \left(\prod_{i=1}^{d} \mathrm{d}u_i \widehat{P}_i(u_i)\right) \delta\left(h - f(u_1, \ldots, u_d)\right) z_0(u_1, \ldots, u_d)^m$$

the mean distributions Q and $\widehat{Q}$ defined as:

$$Q = \int \mathrm{d}P \mathcal{P}^{1\mathrm{RSB}}(P) P, \quad \widehat{Q} = \int \mathrm{d}\widehat{P} \widehat{\mathcal{P}}^{1\mathrm{RSB}}(\widehat{P}) \widehat{P}$$

which must be symmetric probability for **the global spin-flip symmetry** $Q(h) = Q(-h)$ and $\widehat{Q}(u) = \widehat{Q}(-u)$

$$Q(h) = \sum_{d=0}^{\infty} p_d \int \left(\prod_{i=1}^{d} \mathrm{d}u_i \widehat{Q}(u_i)\right) \delta\left(h - f(u_1, \ldots, u_d)\right) \frac{z_0(u_1, \ldots, u_d)}{z_0(0, \ldots, 0)}$$

$$\widehat{Q}(u) = \int \left(\prod_{i=1}^{k-1} \mathrm{d}h_i Q(h_i)\right) \delta\left(u - g(h_1, \ldots, h_{k-1})\right) \frac{\widehat{z}_0(h_1, \ldots, h_{k-1})}{\widehat{z}_0(0, \ldots, 0)}$$

4.2.2 Simplifications for $m = 1$

Thanks to the symmetry of Q and $\hat{Q}$ these are well-normalized, and are related to the original distributions by $Q(h) = \frac{1}{2}(Q_+(h) + Q_-(h))$

$$Q_+^{(t+1)}(h) = \sum_{d=0}^{\infty} p_d \int \left(\prod_{i=1}^{d} \mathrm{d}u_i \, \widehat{Q}_+^{(t)}(u_i) \right) \delta\left(h - f\left(u_1, \ldots, u_d\right)\right)$$

$$\widehat{Q}_+^{(t)}(u) = \sum_{\{\sigma_1\}} \tilde{p}\left(\sigma_1, \ldots, \sigma_{k-1} \mid +\right) \int \left(\prod_{i=1}^{k-1} \mathrm{d}h_i \, Q_+^{(t)}(h_i) \right) \delta\left(u - g\left(\sigma_1 h_1, \ldots, \sigma_{k-1} h_{k-1}\right)\right)$$

where

$$\tilde{p}\left(x_1, \ldots, x_{k-1} \mid +\right) = \frac{\omega\left(x_1, \ldots, x_{k-1}, +\right)}{\sum_{\{x'\}} \omega\left(x_1', \ldots, x_{k-1}', +\right)} = \frac{\sum_{p=0}^{k-1} \omega_p \mathbb{I}\left[\sum_{i=1}^{k-1} x_i = k - 1 - 2p\right]}{\sum_{p=0}^{k-1} \begin{pmatrix} k-1 \\ p \end{pmatrix} \omega_p}$$

RS free entropy

$$\Phi_{\mathrm{RS}} = \sum_{d=0}^{\infty} p_d \int \left(\prod_{i=1}^{d} \mathrm{d}u_i \, \widehat{Q}_+(u_i) \right) \ln \mathcal{Z}_0^{\mathrm{v}}\left(u_1, \ldots, u_d\right) + \alpha \int \left(\prod_{i=1}^{k} \mathrm{d}h_i \, Q_+(h_i) \right) \ln \mathcal{Z}_0^{\mathrm{c}}\left(h_1, \ldots, h_k\right)$$
$$- \alpha k \int \mathrm{d}h \, \mathrm{d}u Q_+(h) \widehat{Q}_+(u) \ln \mathcal{Z}_0^{\mathrm{e}}(h, u)$$

4.3 Tilted and Pinned measures

Tilted measure:

$$f^{\text{tilt}}\left(u_1,\ldots,u_d\right)=\frac{e^{2\gamma(\gamma+z)}\prod_{i=1}^{d}\left(1+u_i\right)-\prod_{i=1}^{d}\left(1-u_i\right)}{e^{2\gamma(\gamma+z)}\prod_{i=1}^{d}\left(1+u_i\right)+\prod_{i=1}^{d}\left(1-u_i\right)},\quad z\sim\mathcal{N}(0,1)$$

$$\mathcal{Z}_0^{\text{v}}\left(u_1,\ldots,u_d\right)=e^{\gamma(\gamma+z)}\prod_{i=1}^{d}\left(\frac{1+u_i}{2}\right)+e^{-\gamma(\gamma+z)}\prod_{i=1}^{d}\left(\frac{1-u_i}{2}\right),\quad z\sim\mathcal{N}(0,1)$$

Pinning measure:

$$f^{\text{pinn}}\left(u_1,\ldots,u_d\right)=\begin{cases}1 & \text{with probability } \theta\\ \frac{\prod_{i=1}^{d}(1+u_i)-\prod_{i=1}^{d}(1-u_i)}{\prod_{i=1}^{d}(1+u_i)+\prod_{i=1}^{d}(1-u_i)} & \text{with probability } 1-\theta\end{cases}$$

$$\mathcal{Z}_0^{\text{v}}\left(u_1,\ldots,u_d\right)=\begin{cases}\prod_{i=1}^{d}\left(\frac{1+u_i}{2}\right) & \text{with probability } \theta\\ \prod_{i=1}^{d}\left(\frac{1+u_i}{2}\right)+\prod_{i=1}^{d}\left(\frac{1-u_i}{2}\right) & \text{with probability } 1-\theta\end{cases}$$